Wendian **Luo**

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Research interests: Federated Learning, Edge Intelligence, and Multimodal Models

Education

Sichuan University, College of Computer Science — Chengdu, China

M.Eng. in Computer Technology — Sep. 2024 – Jun. 2027 (Expected)

- GPA: 3.65/4.00; First-Class Graduate Academic Scholarship.

Sichuan University, College of Computer Science — Chengdu, China

B.Eng. in Computer Science and Technology — Sep. 2020 – Jun. 2024

- GPA: 3.72/4.00; class rank: 38/318 (top 12%).
- Ranked 8/318 in selection for recommendation-based postgraduate admission.
- Outstanding Graduate of Sichuan Province, 2024.

Research Experience

FedSAE: Energy-Efficient Federated Learning for Edge Devices — Sichuan University

First Author and Researcher — 2025

- Designed a semi-asynchronous end–edge–cloud federated prototype learning framework for heterogeneous data, models, and devices, exchanging class-level feature statistics and calibrating a global classifier with virtual features.
- Integrated client-side dynamic voltage and frequency scaling (**DVFS**) with federated training and validated the framework on **NVIDIA Jetson TX2**, demonstrating improved energy efficiency and predictive performance.

LPSAFS: Low-Power Scheduling with Segmented Frequency Scaling — Sichuan University

First Author and Researcher — 2024

- Designed a minimum-slack allocation strategy for periodic real-time tasks under discrete CPU frequencies, balancing energy consumption, task deadlines, and reliability through constrained frequency scaling.
- Implemented and evaluated the scheduler on a **Kendryte K210 RISC-V** platform with **FreeRTOS and EDF scheduling**, integrating low-power control into real-time task execution.

TMSC: Zero-Shot Industrial Anomaly Detection — Pattern Recognition

First Author and Researcher — 2026 | Accepted

- Developed a DINOv3/CLIP-based anomaly detection framework with cross-modal attention and tri-regional semantic calibration; paper accepted by *Pattern Recognition*.

Publications

Journal metrics: 2025 Journal Impact Factor (IF), 2026 JCR release.

Journal Articles

1. **Wendian Luo**, Tong Yu, Shengxin Dai, Bing Guo, Xuesen Lin, and Yanglin Pu, "Semi-Asynchronous Energy-Efficient Federated Prototype Learning for End-Edge-Cloud Architectures," ***Future Generation Computer Systems***, vol. 179, Art. no. 108351, 2026. **IF: 5.9.** doi:10.1016/j.future.2025.108351
2. Shengxin Dai, Zheng Luo, **Wendian Luo**, Siyi Wang, Cheng Dai, Bing Guo, and Xiaokang Zhou, "Energy-Efficient Inference With Software–Hardware Co-Design for Sustainable Artificial Intelligence of Things," ***IEEE Internet of Things Journal***, vol. 11, no. 24, pp. 39170–39182, 2024. **IF: 8.7.** doi:10.1109/JIOT.2024.3482288
3. **Wendian Luo**, Shengxin Dai, Cheng Dai, Bing Guo, Sherif Moussa, and Mubarak Alrashoud, "BDTest: A Diversity-Oriented Test Case Generation Framework for Deep Neural Networks in 6G-IOT," ***IEEE Internet of Things Journal***, vol. 13, no. 5, pp. 8190–8203, 2026. **IF: 8.7.** doi:10.1109/JIOT.2025.3616358
4. **Wendian Luo**, Tong Yu, Junnan Li, Shengxin Dai, Bing Guo, and Zhuohang Jiang, "TMSC: A Tri-Regional Mask-Guided Semantic Calibration Framework for Zero-Shot Anomaly Detection," ***Pattern Recognition***, 2026 (Accepted). **IF: 9.1.**

Conference Papers

1. **Wendian Luo**, Shengxin Dai, and Bing Guo, "A Low Power Scheduling Algorithm for Segmented Frequency Scaling," in ***2025 IEEE International Symposium on Parallel and Distributed Processing with Applications (ISPA)***, pp. 828–837, 2025. doi:10.1109/ISPA67752.2025.00112

Selected Projects

Key Technologies for Highly Reliable, Low-Power Industrial Control Systems

Completed

Project Contributor | Embedded Low-Power Scheduling

DEC–SCU Joint Innovation Institute (Sichuan University allocation: RMB 200,000)

- The project investigated energy-efficient operation and reliable execution for resource-constrained industrial control systems, with low-power real-time scheduling as a core research task.
- Implemented **LPSAFS on K210 under FreeRTOS**, supporting discrete CPU frequency states and reclaiming task slack to reduce energy use while limiting aggressive downscaling and maintaining deadline and reliability constraints.

Edge-Side Large Models for Next-Generation Mobile Communications

Mid-term Stage

Project Contributor | End–Edge–Cloud Collaborative Training

Sichuan Provincial Major Science and Technology Project (Sichuan University allocation: RMB 750,000)

- The project develops edge-side models and heterogeneous terminals for equipment monitoring in transportation, hydropower, and aviation, supported by cloud–edge–device coordination.
- Developed **FedSAE** for collaborative training across heterogeneous devices, combining prototype learning and DVFS to improve training energy efficiency; published the resulting first-author paper in *Future Generation Computer Systems*.

Interactive Multimodal Audiovisual Generation and Presentation

Early Stage

Project Contributor | Proposal Coordination and Technical Planning

Sichuan Provincial Key Research and Development Project (Sichuan University allocation: RMB 1.5 million)

- The project targets multimodal content generation, interactive presentation across terminals, and text and video content safety review, with planned development of an AI video-stream analysis terminal.
- Consolidated contributions to the project proposal, drafted the research plan and technical approach for **text and video content safety review**, and prepared the project review presentation.

Professional Experience

Media Intelligence Laboratory, Chengdu Sobey

Chengdu, China

Large Language Model Algorithm R&D Intern Jul. 2024 – Sep. 2024

- Constructed and cleaned English instruction-tuning datasets for Story-to-Scene generation, scene keyword extraction, and shot-level visual description tasks; developed scripts for translation, provenance tracing, deduplication, and format validation.
- Fine-tuned and evaluated Qwen-72B with LLaMA-Factory and LoRA on an eight-GPU NVIDIA A800 server.
- Deployed and integrated inference services using vLLM, Flask, and HTTP APIs, and completed API documentation and project handover.

Technical Skills

Federated & Edge AI	Federated prototype learning, end–edge–cloud collaboration, energy-aware training, NVIDIA Jetson TX2/Orin/AGX/Thor
Embedded Systems	C, RISC-V/Kendryte K210, FreeRTOS, EDF real-time scheduling, discrete frequency scaling, DVFS
ML & Deployment	Python, PyTorch, model deployment, vLLM, Ollama, LoRA, CLIP, DINOv3
Development Tools	Linux, Git, LaTeX
Languages	English: CET-4 and CET-6; IELTS 6.5 overall (minimum band 5.5)

Honors & Awards

2027	**Outstanding Master's Graduate of Sichuan Province**, Sichuan Province	China
2024	**Outstanding Graduate of Sichuan Province**, Sichuan Province	China
2024–27	**First-Class Graduate Academic Scholarship**, Sichuan University	Chengdu
2023	**National Second Prize**, 12th China Software Cup	China
2023	**First Prize, Sichuan Division**, Chinese College Students Computer Design Competition	Sichuan
2020–24	**Merit and Academic Scholarships**, Sichuan University	Chengdu